P101 · Unit 7 · Quick reference

Physics, space & scientific thinking
Thermal transfers and testing a motion model
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Recall **Temperature**	T / K $= \theta$ / °C + 273.15. A change of 1 °C equals a change of 1 K. Do not add the offset to a temperature interval.
Understand **State and transfer**	Internal energy belongs to the system's microscopic state. Heat is transfer caused by a temperature difference; work is another transfer mechanism. Neither is stored "heat". Choose the system boundary.
Recall **First law**	$\Delta E_{\text{int}} = Q + W_{\text{on}} = Q - W_{\text{by}}$ when other stored-energy changes are negligible. Inward heat and work on the system are positive.
Understand **Heat capacity**	$\Delta E_{\text{int}} = C\Delta T = mc\Delta T$, with C in J/K and c in J/(kg K). Use an appropriate approximately constant capacity, fixed phase and sufficiently uniform temperature; include other energy changes when relevant.
Recall **Isolated equilibrium**	$T_f = \frac{C_1T_1 + C_2T_2}{C_1 + C_2}$. Positive capacities; common final temperature; no external heat, work or phase change. Include a vessel as another term if needed. The endpoint lies between the starting levels.
Understand **Microscopic model**	For a dilute ideal monatomic gas, temperature tracks mean translational kinetic energy per particle. Total internal energy also depends on particle number. This is not a universal microscopic formula for all materials.

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Recall **Rates of motion**	$v = \frac{\mathrm{d}x}{\mathrm{d}t}$, $a = \frac{\mathrm{d}v}{\mathrm{d}t}$. For $x = x_0 + v_0 t + \frac{1}{2}at^2$, constant a gives $v = v_0 + at$. A zero of velocity is a turn only if velocity changes sign.
Understand **Speed and signs**	Speed is $\lvert v \rvert$. Away from $v = 0$, equal signs of v and a mean increasing speed; opposite signs mean decreasing speed. Displacement and distance are different.
Consult **Centred estimate**	$v(t) \approx \frac{x(t+h) - x(t-h)}{2h}$, with $h > 0$ and both sample times inside the domain. Reducing h can reduce approximation error while amplifying rounded-input effects.
Consult **Stopping models**	For initial speed $v > 0$, constant braking magnitude $b > 0$: $t_A = \frac{v}{b}$, $d_A = \frac{v^2}{2b}$. A constant-speed delay $\tau \geq 0$ adds τ to time and $v\tau$ to distance.
Recall **Residual**	$r_i = d_{\text{observed},\,i} - d_{\text{predicted},\,i}$; MAR $= \frac{1}{n}\sum_i \lvert r_i \rvert$. Keep repeated observations. Plot residuals as well as calculating a summary.
Understand **Evidence and units**	1 mph = 0.44704 m/s; 1 ft = 0.3048 m. A benchmark tests implementation. Observations test predictions under documented conditions. A smaller MAR alone does not identify a cause or validate a universal law.

Consult · Python

Use NumPy arrays for the recorded speeds and distances; b > 0 and tau ≥ 0. Inspect residual plots too.

```
v = speed_mph * 0.44704
observed = distance_ft * 0.3048
predicted = v**2 / (2*b) + v*tau
residual = observed - predicted
mar = sum(abs(r) for r in residual) / len(residual)
```

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